13 A number sequence starts 1 2 4

Emma says that the next term is 7

(a) Explain why Emma may be correct.

Differences are +1, +2, then +3; 4 + 3 = 7.

1 mark
Successive differences

(1)

Here are the first four terms of the sequence of triangle numbers.

1 3 6 10

(b) Find the 8th term of this sequence.

Triangle numbers add 2, 3, 4, 5, 6, 7, 8.

Terms: 1, 3, 6, 10, 15, 21, 28, 36.

2 marks total
Continue pattern
8th term 36

(2)

(Total for Question 13 is 3 marks)

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